

GLM (Generalized Linear Models)

Background : The exponential family

$$p(y; \eta(\theta)) = b(y) \exp(\eta(\theta)^T T(y) - a(\eta(\theta))) \quad // \theta = \text{chld distribution's parameter}$$

- $\eta(\theta)$: natural parameter, could be a vector.

- $a(\eta(\theta))$: log partition function, $e^{-a(\eta(\theta))}$ is the constant making p integrate to 1.

$$\cdot \nabla_{\eta(\theta)} a(\eta(\theta)) = E[T(y)]$$

$$\cdot \nabla_{\eta(\theta)}^2 a(\eta(\theta)) = \text{Cov}[T(y)] \quad // \text{ since Cov matrix is PSD, } a \text{ is convex respect to } \eta(\theta).$$

- $b(y)$: base measure

- $T(y)$: sufficient statistic, could be a vector.

should only depend on $\eta(\theta)$

should only depend on y

↳ GLM Assumptions

1. $y | x; w \sim \text{ExponentialFamily}(\eta)$

// η is a function of θ (chld distribution's parameter)

2. $\eta = w^T x^{(i)}$

3. $h_w(x^{(i)})$ (model's prediction) = $E[T(y^{(i)}) | x^{(i)}] = E_{\eta}[T(y^{(i)})]$

* This turns out to be $a'(\eta(\theta))$ // Canonical response function $g(\eta)$, that takes η to the prediction value (Inverse: "link")

⊕ Since a is convex to η and $\eta = w^T x$ is affine to w ,

NLL is convex to w .

We want to find $\underset{w}{\operatorname{argmax}} \prod_{i=1}^n p(y^{(i)} | \eta^{(i)})$

$$= \underset{w}{\operatorname{argmax}} \log \prod_{i=1}^n p(y^{(i)} | \eta^{(i)}) = \underset{w}{\operatorname{argmax}} \sum_{i=1}^n \log p(y^{(i)} | \eta^{(i)})$$

$$= \underset{w}{\operatorname{argmin}} \sum_{i=1}^n -\log p(y^{(i)} | \eta^{(i)})$$

$$= \underset{w}{\operatorname{argmin}} -\sum_{i=1}^n \log [b(y^{(i)}) \cdot \exp(\eta^{(i)T} T(y^{(i)}) - a(\eta^{(i)}))]$$

$$= \underset{w}{\operatorname{argmin}} -\sum_{i=1}^n \log b(y^{(i)}) + \eta^{(i)T} T(y^{(i)}) - a(\eta^{(i)})$$

$$= \underset{w}{\operatorname{argmin}} \sum_{i=1}^n a(\eta^{(i)}) - \eta^{(i)T} T(y^{(i)}) \quad // \text{ Loss function}$$

$$\nabla_w [a(\eta^{(i)}) - \eta^{(i)T} T(y^{(i)})]$$

$$= \sum_{i=1}^n a'(\eta^{(i)}) \cdot x^{(i)} - T(y^{(i)}) \cdot x^{(i)}$$

$$= \sum_{i=1}^n (E[T(y^{(i)})] - T(y^{(i)})) \cdot x^{(i)} \quad // \text{ Gradient}$$

Modeling $p(y^{(i)} | x^{(i)})$ as a Gaussian distribution with variance 1

$$y^{(i)} | x^{(i)} \sim \text{Exp Fam}(\eta) = \mathcal{N}(\mu, 1)$$

$$\mathcal{N}(\mu, 1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y^{(i)} - \mu)^2}{2}} = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{y^{(i)2}}{2} + y^{(i)}\mu - \frac{\mu^2}{2}} = \left(\frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{y^{(i)2}}{2}} \right) e^{y^{(i)}\mu - \frac{\mu^2}{2}}$$

$$\begin{cases} \cdot b(y^{(i)}) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^{(i)2}}{2}} \\ \cdot T(y^{(i)}) = y^{(i)} \\ \cdot \eta = \mu \\ \cdot a(\eta) = \frac{\mu^2}{2} = \frac{\eta^2}{2} \end{cases}$$

$$\Rightarrow \text{Prediction: } \nabla_{\eta} a(\eta) = \eta = w^T x^{(i)}$$

$$\text{Loss function: } \sum_{i=1}^n \left(\frac{\mu^2}{2} - \mu y^{(i)} \right) = \sum_{i=1}^n \left[\frac{1}{2} (w^T x^{(i)} - y^{(i)})^2 - \frac{1}{2} y^{(i)2} \right]$$

$$\text{Gradient: } \sum_{i=1}^n (w^T x^{(i)} - y^{(i)}) x^{(i)}$$

Modeling $p(y^{(i)} | x^{(i)})$ as a Bernoulli distribution

$$y^{(i)} | x^{(i)} \sim \text{Exp Fam}(\eta) = \text{Bernoulli}(\phi) = \phi^{y^{(i)}} (1-\phi)^{(1-y^{(i)})}$$

$$\begin{aligned} \phi^{y^{(i)}} (1-\phi)^{(1-y^{(i)})} &= \exp(\log(\phi^{y^{(i)}} (1-\phi)^{(1-y^{(i)})}) = \exp(y^{(i)} \log \phi + (1-y^{(i)}) \log(1-\phi)) \\ &= \exp\left(y^{(i)} \log \frac{\phi}{1-\phi} - (-\log(1-\phi))\right) \end{aligned}$$

$$\begin{cases} b(y^{(i)}) = 1 \\ T(y^{(i)}) = y^{(i)} \\ \eta = \log \frac{\phi}{1-\phi} \\ a(\eta) = -\log(1-\phi) = \log(1+e^\eta) \end{cases}$$

$$\begin{aligned} \text{prediction: } \nabla_{\eta} a(\eta) &= \nabla_{\eta} -\log\left(\frac{e^{-\eta}}{1+e^{-\eta}}\right) = \nabla_{\eta} \log(1+e^{\eta}) = \frac{e^{\eta}}{1+e^{\eta}} = \frac{1}{1+e^{-\eta}} \\ &= \underbrace{\text{sigmoid}(\eta)} \end{aligned}$$

$$\left(\eta = \log \frac{\phi}{1-\phi} \leadsto \phi = \frac{1}{1+e^{-\eta}} \right)$$

$$\begin{aligned} \text{Loss function: } &\sum_{i=1}^n \left(-\log(1-\phi) - \left(\log \frac{\phi}{1-\phi} \right) y^{(i)} \right) \\ &= \sum_{i=1}^n y^{(i)} \cdot -\log \phi + (1-y^{(i)}) \cdot -\log(1-\phi) \end{aligned}$$

$$\text{Gradient: } \sum_{i=1}^n (\sigma(W^T x^{(i)}) - y^{(i)}) x^{(i)}$$

Modeling $p(y^{(i)} | x^{(i)})$ as a Multinomial distribution

$$y^{(i)} | x^{(i)} \sim \text{ExpFam}(\eta) = \text{Multinomial}(\underbrace{\phi_1, \dots, \phi_k}_{\text{sum} = 1}) = \prod_{j=1}^k \phi_j^{1\{y^{(i)}=j\}}$$

$$\prod_{j=1}^k \phi_j^{1\{y^{(i)}=j\}} = \exp\left(\log \prod_{j=1}^k \phi_j^{1\{y^{(i)}=j\}}\right) = \exp\left(\sum_{j=1}^k 1\{y^{(i)}=j\} \log \phi_j\right)$$

$$1\{y^{(i)}=k\} = 1 - \sum_{j=1}^{k-1} 1\{y^{(i)}=j\} \quad \Rightarrow$$

$$= \exp\left(\sum_{j=1}^{k-1} 1\{y^{(i)}=j\} \log \phi_j + \left(1 - \sum_{j=1}^{k-1} 1\{y^{(i)}=j\}\right) \log \phi_k\right)$$

$$= \exp\left(\sum_{j=1}^{k-1} 1\{y^{(i)}=j\} \log \frac{\phi_j}{\phi_k} - (-\log \phi_k)\right)$$

$$\left(\begin{array}{l} \cdot b(y^{(i)}) = 1 \\ \cdot T(y^{(i)})_j = 1\{y^{(i)}=j\} \\ \cdot \eta_j = \log \frac{\phi_j}{\phi_k} \quad (= w_j^T x^{(i)}) \\ \eta_k = 0 \\ \cdot a(\eta) = -\log \phi_k = \log\left(1 + \sum_{j=1}^{k-1} e^{\eta_j}\right) \end{array} \right)$$

prediction: $\frac{\partial a(\eta)}{\partial \eta_L} = \frac{\partial}{\partial \eta_L} \log\left(1 + \sum_{j=1}^{k-1} e^{\eta_j}\right) = \frac{e^{\eta_L}}{1 + \sum_{j=1}^{k-1} e^{\eta_j}} = \phi_L = \text{softmax}(\eta)_L$

$\Rightarrow \text{softmax}(Wx^{(i)})$ $// L = 1, 2, \dots, k-1$

Loss function: $\sum_{i=1}^n \left(\log\left(1 + \sum_{j=1}^{k-1} e^{\eta_j^{(i)}}\right) - \sum_{j=1}^{k-1} \eta_j^{(i)} 1\{y^{(i)}=j\} \right) = \sum_{i=1}^n -\log \phi_{y^{(i)}}^{(i)}$

Gradient: $\sum_{i=1}^n (\text{softmax}(Wx^{(i)})_L - 1\{y^{(i)}=L\}) x^{(i)} \quad // L = 1, 2, \dots, k-1$

Modeling $p(y^{(i)} | x^{(i)})$ as a Poisson distribution

$$y^{(i)} | x^{(i)} \sim \text{ExpFam}(\eta) = \text{Poisson}(\lambda) = \frac{e^{-\lambda} \lambda^{y^{(i)}}}{y^{(i)}!}$$

$$\frac{e^{-\lambda} \lambda^{y^{(i)}}}{y^{(i)}!} = \exp\left(\log\left(\frac{e^{-\lambda} \lambda^{y^{(i)}}}{y^{(i)}!}\right)\right) = \exp(-\lambda + y^{(i)} \log \lambda - \log y^{(i)}!)$$

$$b(y^{(i)}) = \frac{1}{y^{(i)}!}$$

$$T(y^{(i)}) = y^{(i)}$$

$$\eta = \log \lambda \quad (= w^T x^{(i)})$$

$$a(\eta) = \lambda = e^\eta$$

$$\text{prediction: } \nabla_\eta a(\eta) = \nabla_\eta e^\eta = e^\eta$$

$$\text{Loss function: } \sum_{i=1}^n \lambda - \log \lambda \cdot y^{(i)} = \sum_{i=1}^n e^{\eta^{(i)}} - \eta^{(i)} y^{(i)}$$

$$\text{Gradient: } \sum_{i=1}^n (e^\eta - y^{(i)}) x^{(i)}$$

Modeling $p(y^{(i)} | x^{(i)})$ as a Gamma distribution

$$p(y^{(i)} | x^{(i)}) \sim \text{ExpFam}(\eta(\theta)) = \text{Gamma}(c, d) = \frac{(y^{(i)})^{c-1} e^{-\frac{y^{(i)}}{d}}}{\Gamma(c) d^c}$$

$$\frac{(y^{(i)})^{c-1} e^{-\frac{y^{(i)}}{d}}}{\Gamma(c) d^c} = \exp\left(\log\left(\frac{(y^{(i)})^{c-1} e^{-\frac{y^{(i)}}{d}}}{\Gamma(c) d^c}\right)\right) = \exp\left((c-1) \log y^{(i)} - \frac{y^{(i)}}{d} - \log \Gamma(c) d^c\right)$$

$$\left(\begin{array}{l} b(y^{(i)}) = 1 \\ T(y^{(i)}) = \begin{bmatrix} \log y^{(i)} \\ y^{(i)} \end{bmatrix} \\ \eta(c, d) = \begin{bmatrix} c-1 \\ -\frac{1}{d} \end{bmatrix} \quad \begin{array}{l} = W_1^T x^{(i)} \\ = W_2^T x^{(i)} \end{array} \\ a(\eta(c, d)) = \log \Gamma(c) d^c \end{array} \right.$$

⇒ We usually fix one of the params and proceed